

CAVATICA



Access data, run analyses, and collaborate

ON THE CLOUD

In this session, you will learn...

- ▶ What is CAVATICA, powered by Seven Bridges, and why is it on the Cloud?
- ▶ What can you do on CAVATICA? And why should you work in the cloud?
- ▶ Where to find example resources to get started

Let's take a poll!



How comfortable are you with
the cloud and cloud data?

1



Not
comfortable
*I've used it very little,
or never*

2



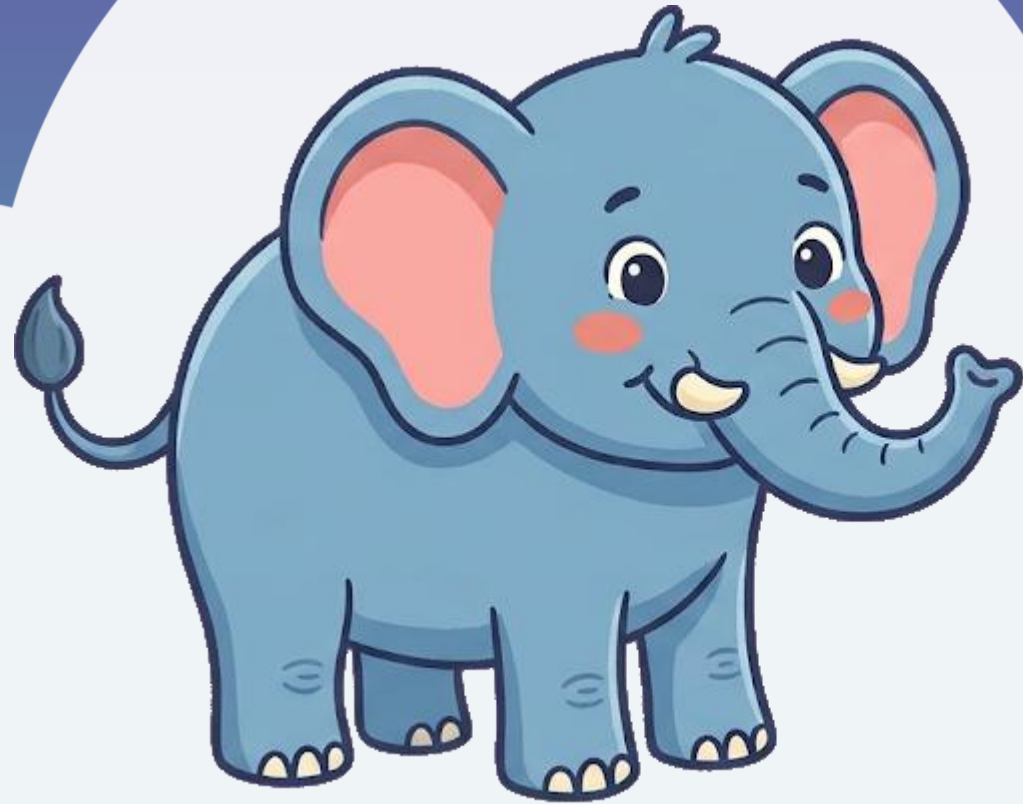
Somewhat
comfortable
I've used it before

3



Very comfortable
I use it all the time

The elephant in the room: Most of you don't need the cloud



Can learning a new cloud-native platform benefit YOU?



Transferable

Data and tools on CAVATICA can move with you. Data and tools on your HPC, not so much.



Data access

When you're ready to go beyond your own data, seamlessly use NIH-hosted datasets.



Community

Hundreds of curated analysis tools and public projects, plus office hours with the experts.

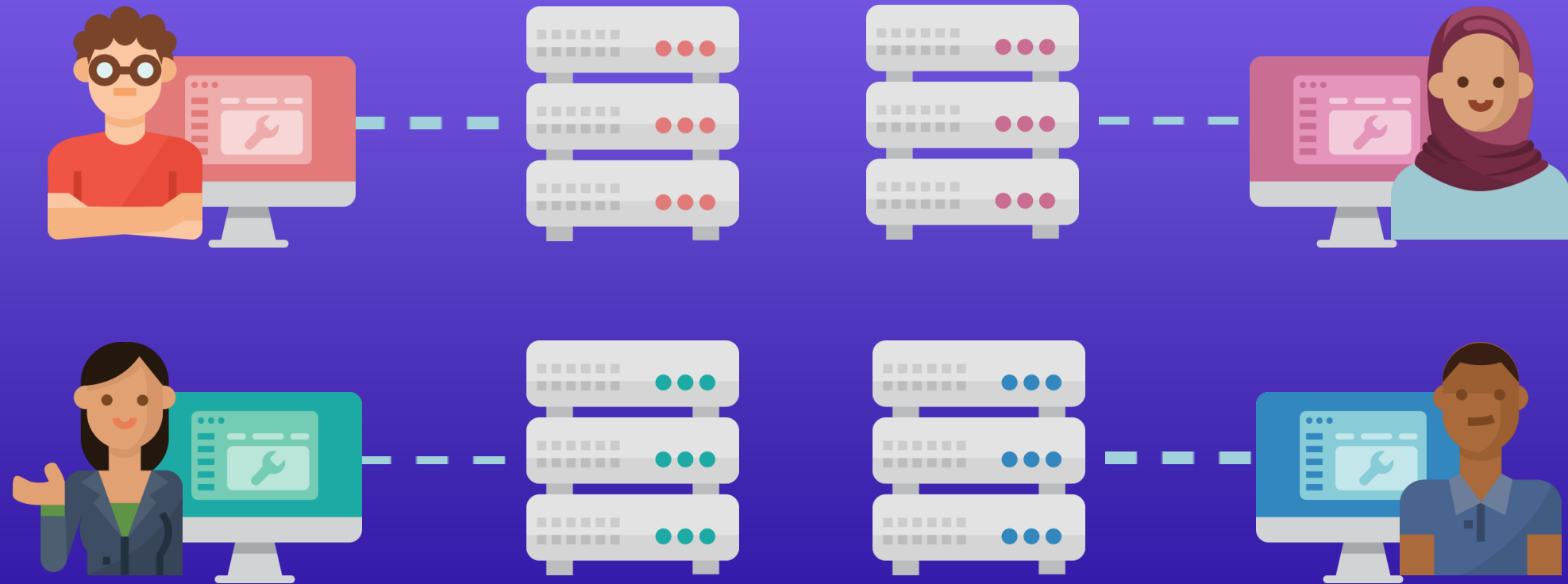


Future proof

Better (and easier) to learn now! NIH is subsidizing the future of biomedical research.

Traditional Approach

Bring data to researchers



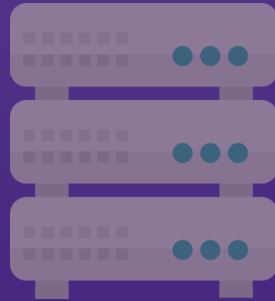
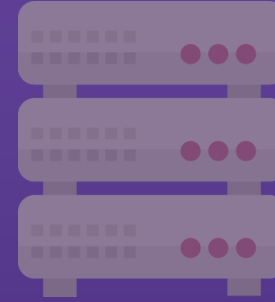
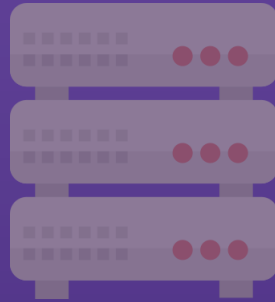
What's wrong with this traditional approach? ?

Data is
copied, not
shared

High storage &
infrastructure
costs

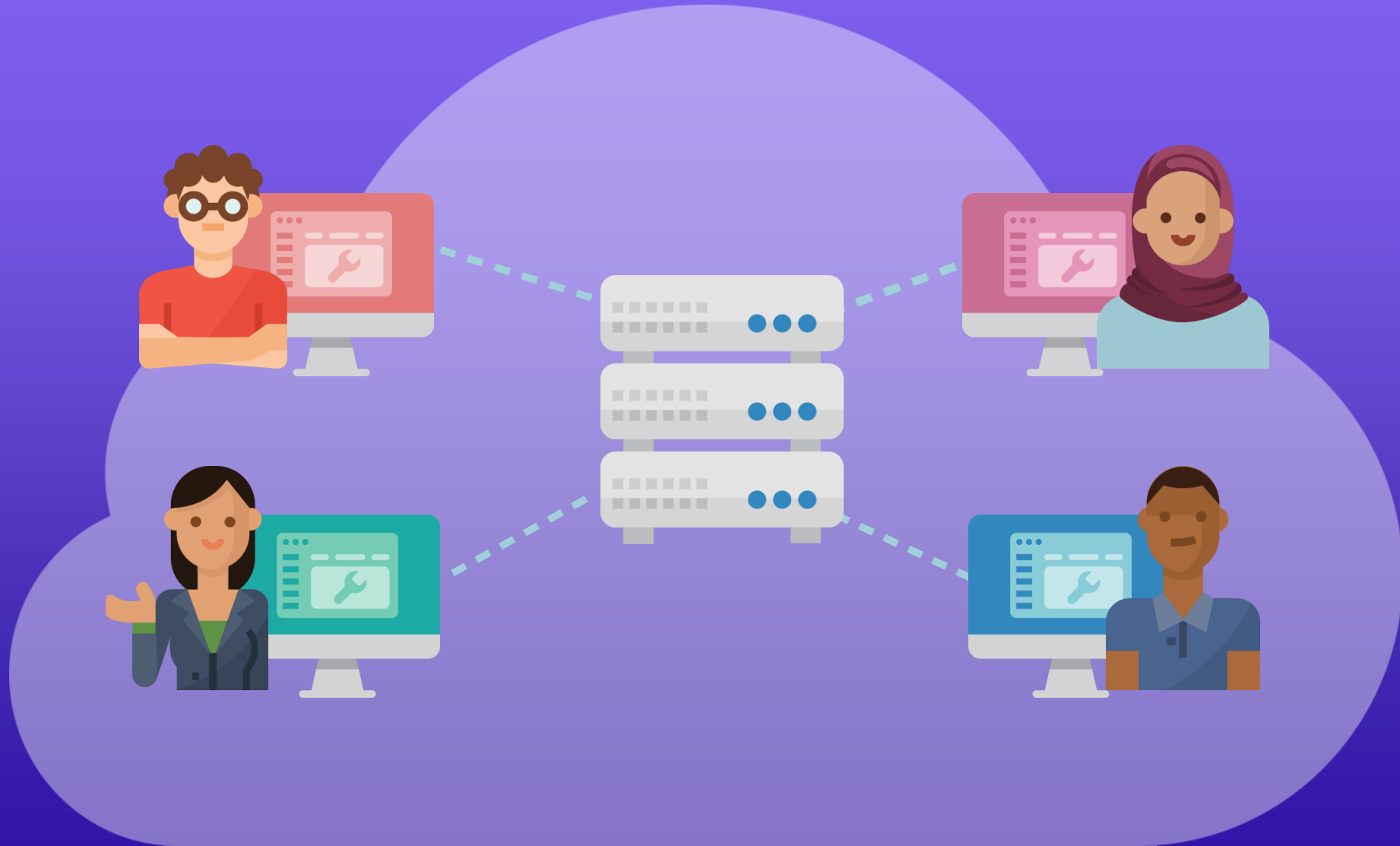
Hard to
share/
reproduce

Security on
individuals



A new cloud-native vision for biomedical research

CAVATICA



A new cloud-native vision for biomedical research

CAVATICA

~~Not
copies, not
shared~~

~~High storage &
infrastructure
costs~~

~~Hard to
share/
reproduce~~

~~Security on
individuals~~

True
immediate
data sharing

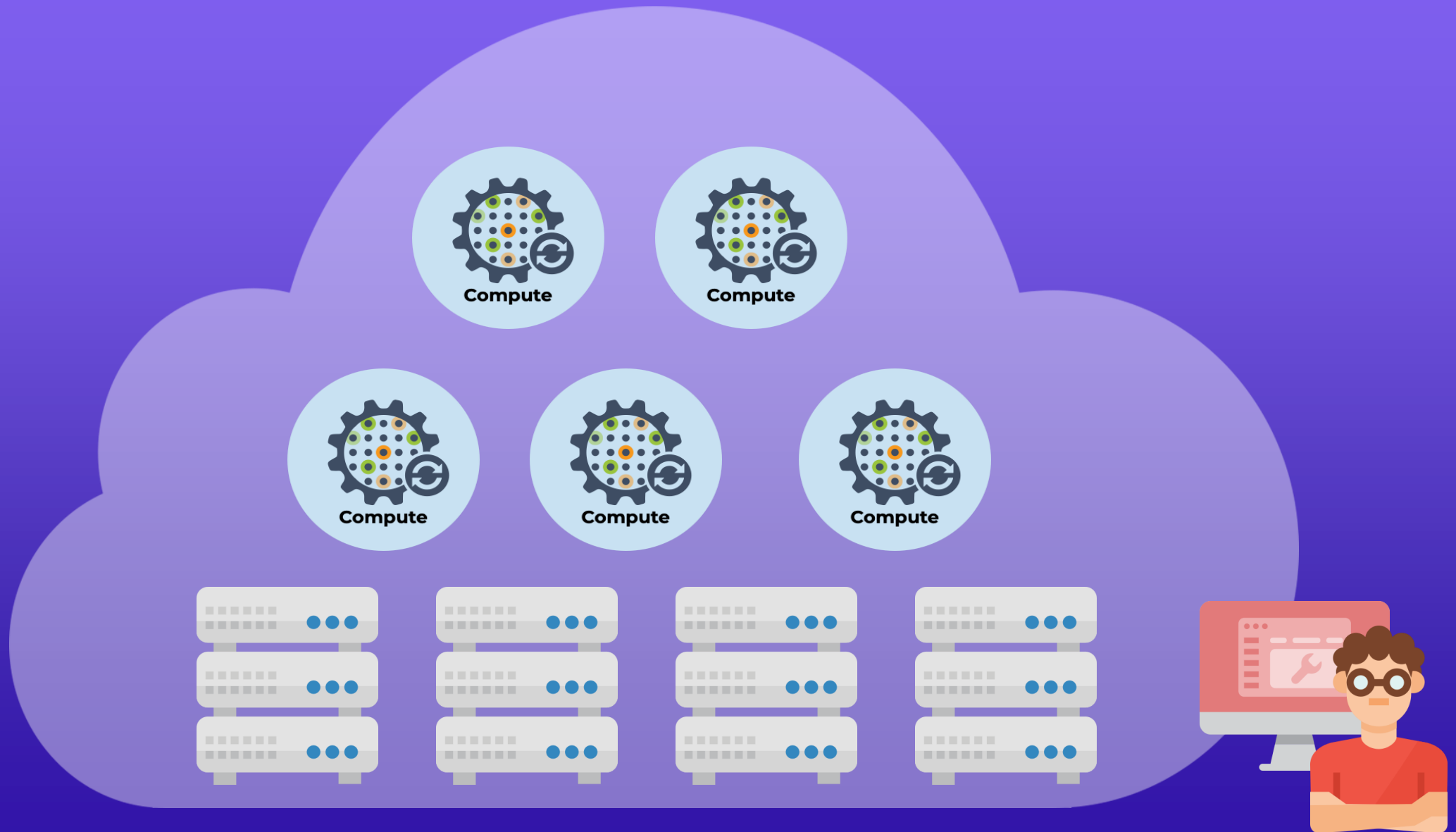
Single storage,
many can
access. Elastic
storage &
compute (pay
for your use)

Shareable,
reproducible
tools and
standardized
environments

Centralized
security



What is data in the cloud??



What is data in the cloud??

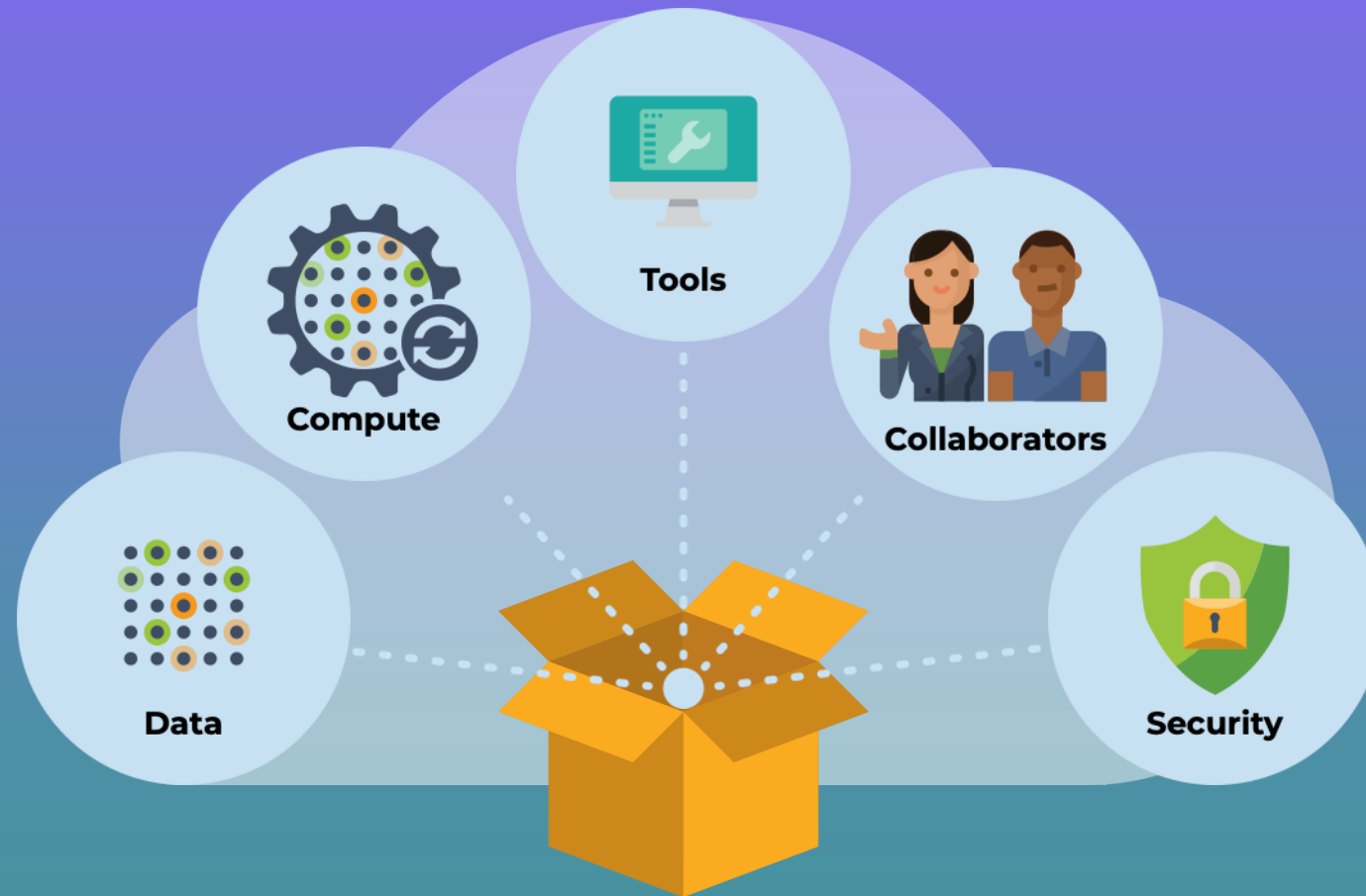
CAVATICA

aws
us-east-1



VELSERA

Migrating to the cloud can be challenging!!

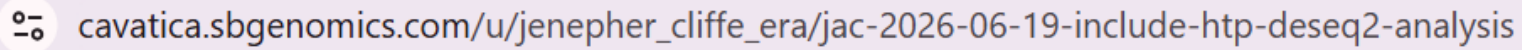


CAVATICA



cavatica.org

CAVATICA



Dashboard Files Apps Tasks Data Studio Interactive Apps **JAC_2026-06-19_INCLUDE_HT...** ⓘ Interactive Browsers Settings Not

INCLUDE HTP DESe

This tutorial project uses high-throughput **data** from the [INCLUDE Data Hub](#), configured for the **cloud** platform. The analysis covers high-throughput differential expression analysis using DESeq2 and Gene Set Enrichment Analysis (GSEA) in R. The analysis specifically focuses on testing for differential gene expression in high-throughput PAXgene Whole Blood RNA samples generated from paired-end PolyA+ globin-depleted libraries.

This project builds computational and data science capacity within the INCLUDE Project and the broader Down Syndrome research

Immediately
& seamlessly
(including
controlled
data)

Easy to use
and find
curated
tools and
resources

As easily as a
Google doc,
with granular
permissions
for control

 Email notifications

jenicliffe
Read

Copy, Write, Execute

 Manage members

Analysis

 Search

Data → Analysis → Result



Projects

Bring data and tools into secure, shareable units



Files

Access public and controlled datasets or bring your own



Data Studios

Interactive analysis sessions for R, Python, and bash.



Apps

Run “tasks” of production-level software & pipelines.

Platform Navigation

The screenshot displays the CAVATICA platform's navigation interface. At the top, a dark navigation bar contains the CAVATICA logo and a series of menu items: Home, Projects, Data, Public Apps, Public Projects, Developer, and Controlled projects. On the right side of this bar are a notification bell icon and a user profile icon labeled 'JC'. Below the navigation bar, the interface is split into two main panels. The left panel, titled 'Projects', features a search bar and a list of project entries, each showing a title, creator, and timestamp. The right panel, titled 'Analysis', has sub-tabs for 'Tasks' and 'Data Studio'. It contains a large message about executions and a section for connecting external data sources like 'Data Catalyst' and 'INCLUDE DRS Server'. Three orange callout boxes with arrows provide additional context: one points to the 'Projects' list, another to the 'Projects' menu item in the top bar, and a third to the user profile icon.

CAVATICA

Home Projects Data Public Apps Public Projects Developer Controlled projects

Projects Search

JAC_2026-06
Created by:jene

2026-06-18_
Created by:jene

2026-06-17_JAC_INCLUDE_HTP_Linear_Regression_Analysis
Created by:jenepher_cliffe_era · Jun 17, 2026, 14:09

INCLUDE_HTP_DESeq2_Analysis
Created by:sevenbridges · Jun 15, 2026, 1:54

INCLUDE_HTP_Linear_Regression_Analysis
Created by:sevenbridges · Jun 15, 2026, 1:48

INCLUDE_HTP_DESeq2_Analysis
Created by:jenepher_cliffe_era · Jun 9, 2026, 16:54

INCLUDE_HTP_Linear_Regression_Analysis
Created by:jenepher_cliffe_era · Jun 9, 2026, 7:29

Analysis Tasks Data Studio

Your executions will appear here.
Before you start, learn more about them.

Data Catalyst
r account to the Gen3 BioData Catalyst platform Not connected

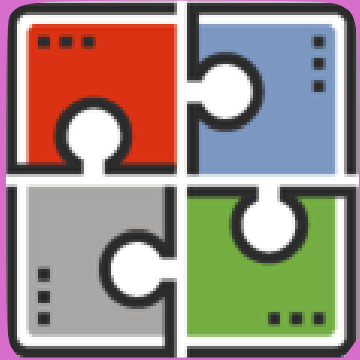
INCLUDE DRS Server
Enable your CAVATICA account to import files from the Connected

This navigation pane is for the entire platform. It lets you find your projects, view available connected data, access public projects (tutorials/walkthroughs), and discover tools & workflows.

Projects can hold all the data and tools for a single study. They can be shared with collaborators and copied for development work.

Access account settings and payment details

Data → Analysis → Result



Projects

Bring data and tools into secure, shareable units



Files

Access public and controlled datasets or bring your own



Data Studios

Interactive analysis sessions for R, Python, and bash.



Apps

Run “tasks” of production-level software & pipelines.

The Project Dashboard

CAVATICA Home Projects ▾ Data ▾ Public Apps ▾ Public Projects ▾

Dashboard Files Apps Tasks Data Studio Interactive Apps

Interactive Browsers Settings Notes

Description Tags

INCLUDE HTP DESeq2

This tutorial project uses high-throughput sequencing data from the [INCLUDE Data Hub](#), configured for use with the [INCLUDE](#) pipeline. The analysis covers high-throughput differential expression analysis using [edgeR](#) and [GSEA](#) in R. The analysis specifically focuses on testing for differential gene expression in high-throughput PAXgene Whole Blood RNA sequencing data from [in-depleted libraries](#).

A project description (markdown) can be edited by project owners, admins, and those with write permission.

For step-by-step instructions to set up and run this analysis, click [here](#).

Members Email notifications

[jenepher_cliffe_era](#) OWNER
Copy, Write, Execute, Admin

[michelle.ingle](#)
Copy, Write, Execute

Manage members

Notes can be used by all members of the project to communicate.

Analysis

Tasks Data Studio

INCLUDE HTP DESeq2 Analysis
Created by: [debarup_sen](#) · Jun 18, 2026 2:03

The Analysis pane will show Tasks and Data Studio sessions that have been run in the project.

This pane supports navigation within a project.

Collaborate securely by adding members with different permissions.

Data → **Analysis** → **Result**



Projects

Bring data and tools into secure, shareable units



Files

Access public and controlled datasets or bring your own



Data Studios

Interactive analysis sessions for R, Python, and bash.



Apps

Run “tasks” of production-level software & pipelines.

Project Files



Root / data

Store and organize files with a familiar directory tree.

Search



Clear filters

Files

analysis_files

data

Linked volumes from AWS or Google)



Name



STAR_2.7.10a_GRCm39_GENCODE38.tar.gz

reference



HTP0669B_MSD



HTP0672A_MSD



HTP0660A_MSD

Primary data files stored in the cloud by NIH are DRS soft links – NIH pays for storage!

Upload from computer, copy from another project, or link from a volume, DRS, or FTP/HTTP.

New Folder

+ Add files ▾

Public Files

Projects

Your Computer

FTP / HTTP

GA4GH Data Repository Service (DRS)

Data Tools



1000G

071026

DRS NA21142.haplotypeCalls.er.raw.g.vcf.gz



1000G

071026

DRS NA21141.haplotypeCalls.er.raw.g.vcf.gz

Size

22.7

911

912

905

910

998

916

Built to work with highly sensitive data

CAVATICA



- ✓ Human genomes
- ✓ Clinical data
- ✓ Other data types



**FISMA
moderate**

Federal Information
Security
Management Act



**FedRAMP
authorized**

Federal Risk and
Authorization
Management Program

Gigabytes of integrated NIH-hosted datasets

[Home](#)[Projects ▾](#)[Data ▾](#)[Public Apps ▾](#)[Public Projects](#)[Developer ▾](#)[Controlled projects](#)

JC

Projects

SB QuickStart

Created by: [jenepher_cliffe_era](#) · Jun 30, 2026, 11:49

JAC_2026-06-19_INCLUDE_HTP_DESeq2_Analysis

Created by: [jenepher_cliffe_era](#) · Jun 19, 2026, 11:41

2026-06-18_INCLUDE_HTP_Linear_Regression_Analysis_JAC

Created by: [jenepher_cliffe_era](#) · Jun 18, 2026, 9:30

New Copy of INCLUDE_HTP_DESeq2_Analysis

Created by: [debarup_sen](#) · Jun 18, 2026, 2:03

New Copy of INCLUDE_HTP_Linear_Regression_Analysis

Created by: [debarup_sen](#) · Jun 18, 2026, 2:03

2026-06-17_JAC_INCLUDE_HTP_Linear_Regression_Analysis

Created by: [jenepher_cliffe_era](#) · Jun 17, 2026, 14:09

INCLUDE_HTP_DESeq2_Analysis

Created by: [sevenbridges](#) · Jun 15, 2026, 1:54

INCLUDE_HTP_Linear_Regression_Analysis

Created by: [sevenbridges](#) · Jun 15, 2026, 1:48

Analysis

[Tasks](#)[Data Studio](#)**COMPLETED****calculate-average_hardcoded run - 06-30-26 16:23:07**

Project: [jenepher_cliffe_era/sb-quickstart](#) · Submitted by: [jenepher_cliffe_era](#) · Jun 30, 2026 12:23



External connections



GEN3 BioData

Connect your account to the [GEN3 BioData Catalyst](#) platform to import data.

Not connected

INCLUDE DRS Server

Enable your CAVATICA account to import files from the INCLUDE DRS Server.

Connected

Kids First FHIR Server

The Kids First Data Resource Centre (Kids First DRC) provides comprehensive genetic and clinical data to support research on

Not connected

When you're ready to go beyond your own data, use NIH-hosted datasets.

Two modes of analysis

Data → **Analysis** → Result



Projects

Bring data and tools into secure, shareable units



Files

Access public and controlled datasets or bring your own



Data Studios

Interactive analysis sessions for R, Python, and bash.



Apps

Production-level software & pipelines.

Interactive Analysis (Data Studio)

CAVATICA

← INCLUDE HTP DESeq2 Analysis

Stop Help ▾

File Edit Code View Plots Session Build Debug Profile Tools Help

rstudio

Addins ▾

HTP_RNASeq_DESeq2

Use familiar environments like JupyterLab, Rstudio, or bash.

```
230 inner_join(globins) %>%
231 inner_join(meta_data) |>
232 mutate(Sampleid = fct_relevel(Sampleid, (gene_percents |
233 ggplot(aes(Sampleid, percent_total, fill = gene_name)) +
234 geom_col() +
235 facet_wrap(~ Karyotype, scale = "free_x") +
236 theme(axis.text.x = element_text(angle = 45, hjust = 1))
237 labs(title = "Globin mRNAs - percentage of total reads")
238 #
239
240
241 # 2. Filter/re
242
```

Run an interactive analysis for data exploration and visualization right in your project

Console Terminal x Back

R 4.4.0 · ~/include-http-c

```
+ mutate(Sampleid = fct_relevel(Sampleid, (gene_percents |> filt
er(Geneid == "ENSG00000244734.4") |> arrange(-percent_total))$Sam
pleid)) |>
+ ggplot(aes(Sampleid, percent_total, fill = gene_name)) +
+ geom_col() +
+ facet_wrap(~ Karyotype, scale = "free_x") +
```

Environment History

Import Data

R ▾ Global Environm

globins 6

meta_data 46

rpkms_data 66

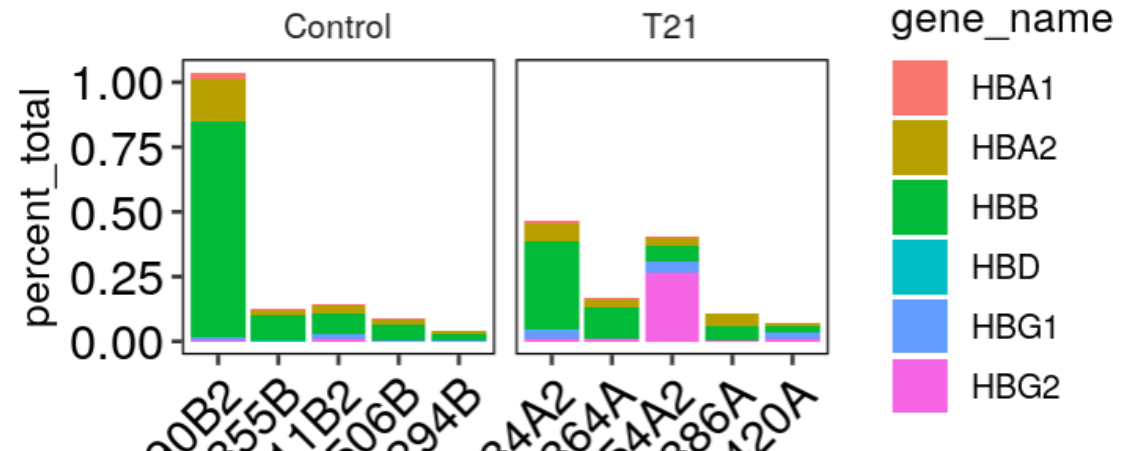
rpkms data lo... 24264800 obs. of 3 variables

Files Plots Packages Help Viewer Presentation

Zoom Export

Charges are based on instance type (many options available) and total session duration. Auto-pause keeps costs under control.

Globin mRNAs - percentage of total reads



Batch jobs for processing

CAVATICA

Data → **Analysis** → Result



Projects

Bring data and tools into secure, shareable units



Files

Access public and controlled datasets or bring your own



Data Studios

Interactive analysis sessions for R, Python, and SAS.



Apps

Automated, production-level software & pipelines.

What is an app?

App

A set of instructions/
multiple connected steps
configured in the platform to
be executed in batch mode.

Common Workflow Language (CWL) or Nextflow

Human-readable language
for data-processing.

```
TGCATTCAGTC
ACGTAAGTCAG
TGCATTCAGTC
ACGTAAGTCAG
TGCATTCAGTC
ACGTAAGTCAG
TGCATTCAGTC
ACGTAAGTCAG
```

Data



Analysis



Result

Develop and modify workflows with a GUI



COPY

RNA-seq alignment - STAR

Version-control for reproducibility

Revision 0 ▾

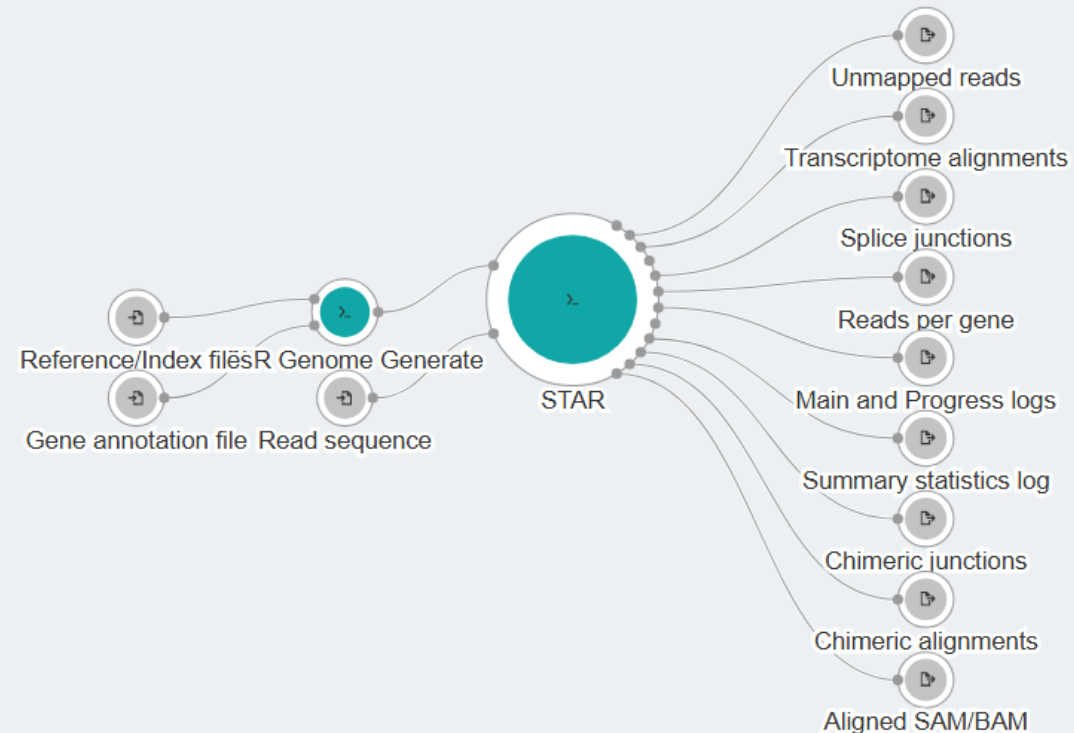
Edit

▶ Run

Copy of RNA-seq alignment - STAR (Revision 2), by jene

2026 08:47

Edit workflows with a graphic user interface – no coding required



Getting Started Resources

Videos

CAVATICA
onboarding on
YouTube



Onboarding

Curated tutorials
and templates
(Public Projects) to
adapt to your data.

Vetted and tested
apps and tools

Support

24/7 help or office
hours twice/week





Home

Projects ▾

Data ▾

Public Apps ▾

Public Projects

Developer ▾

Controlled projects



JC

Dashboard

Files

Apps

Tasks

Data Studio

Search

Analysis Name

Status

INCLUDE HTP DESeq2 Analysis

SAVED

Need help?

Learn from the documentation below.

[About Data Studio](#)[Data Studio analysis editor](#)[Run a Data Studio analysis](#)[Data Studio analysis charging](#)[Data Studio environments and libraries](#)Not finding what you need? Visit our [Knowledge Center](#).**Contact our support**

Describe your issue or share your ideas

Send

...



Interactive Browsers

Settings

Notes

+ Add analysis

Create new analysis



Created on

Action

Jun. 19, 2026 11:41

Start

Contact support
from anywhere on
the platform

Let's take a tour of CAVATICA!



Before continuing...

Note that CAVATICA is most compatible with Google Chrome.

OK

Sign up for cloud credits from **INCLUDE!**

- ✓ \$100 Pilot credits to try it out
- ✓ \$1k NIH awards for research projects



Dedicated DS3 2026 Office Hour

Join us!

Week of July 27, 2027
Exact date/time TBD



<https://qr.link/z7gCnt>

Contact: jenepher.cliffe@velsera.com